

Total Lagrangian Finite Element Analysis of Solid and Porous Hyperelastic (Gent) Specimens using 4-Node Quadrilateral and 6-Node Triangular Elements

ME676: Nonlinear Finite Element Methods
Term Project Report

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Abstract

This report documents the development and application of a total-Lagrangian, displacement-based nonlinear finite element (FE) code for two-dimensional, plane-strain finite elasticity of compressible hyperelastic solids. The Gent strain-energy function is used to model a rubber-like (elastomeric) material, and the resulting Newton–Raphson code is implemented for two isoparametric element families – the 4-node bilinear quadrilateral and the 6-node quadratic triangle. Both full (Gauss–Legendre) integration and a selective reduced integration (SRI) scheme, in which the volumetric part of the stress is under-integrated, are implemented in order to study and mitigate volumetric locking in the nearly incompressible limit. The code is first validated against the classical Cook’s membrane benchmark problem using a compressible neo-Hookean limit of the Gent model, and results are compared with the mixed finite-element results of Brink and Stein (1996). The validated code is then used to analyse the large tensile deformation of a dog-bone shaped elastomeric specimen in two configurations: a fully *solid* specimen and a *porous* specimen containing five circular voids. Stress contours, load–displacement response and the influence of the integration scheme are compared across all eight combinations of (solid/porous) $\times$ (4-node/6-node) $\times$ (full/selective-reduced integration).

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1 Introduction

Elastomeric (rubber-like) materials undergo large, reversible deformations and are almost incompressible, which makes their finite element analysis considerably more demanding than that of linear elastic solids. Two distinct sources of numerical difficulty arise: (i) the material and geometric *nonlinearity* associated with finite strains, which requires an incremental-iterative (Newton–Raphson) solution procedure built on a consistent linearization of the weak form, and (ii) *volumetric locking*, an artificial over-stiffening of low-order displacement elements when the bulk modulus is much larger than the shear modulus (the nearly incompressible limit).

The objective of this project is to:

- (i) Formulate and implement a total Lagrangian (TL), displacement-based finite element code for plane-strain finite elasticity using a compressible **Gent** hyperelastic material model;
- (ii) Implement two element types – a 4-node bilinear quadrilateral (4NR) and a 6-node quadratic triangle (6NT) – together with both full integration and a selective reduced integration (SRI) scheme;
- (iii) Validate the implementation against the well-known **Cook’s membrane** benchmark, comparing against published mixed finite-element results [1];
- (iv) Apply the validated code to the large-strain tensile response of a **solid** and a **porous** (five-hole) dog-bone elastomeric specimen, and study the effect of element type and integration scheme on the predicted stresses and on volumetric locking.

2 Theoretical Formulation

2.1 Kinematics

Let $\mathbf{X}$ denote the position of a material point in the reference (undeformed) configuration and $\mathbf{x} = \mathbf{X} + \mathbf{u}(\mathbf{X})$ its position in the current (deformed) configuration, where $\mathbf{u}$ is the displacement field. The deformation gradient is

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u}, \quad J = \det \mathbf{F}, \quad (1)$$

with $J > 0$ enforcing local invertibility of the deformation. The first strain invariant of the right Cauchy–Green tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ is

$$I_1 = \text{tr}(\mathbf{C}) = \text{tr}(\mathbf{F}^T \mathbf{F}). \quad (2)$$

2.2 Gent Hyperelastic Material Model

The material is modelled using the (compressible) **Gent** strain-energy function, which augments the classical neo-Hookean model with a finite-extensibility (strain-locking) parameter α and a volumetric penalty term:

$$\Omega(\mathbf{F}) = -\frac{\mu\alpha}{2} \ln \left[1 - \frac{I_1 - 3}{\alpha} \right] - \mu \ln J + \frac{G'}{2} (J - 1)^2, \quad (3)$$

where μ is the small-strain shear modulus, G' is a penalty (bulk-type) modulus enforcing near incompressibility, and α controls the limiting chain stretch (as $\alpha \rightarrow \infty$, Eq. (3) reduces to the standard compressible neo-Hookean energy $\Omega = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{G'}{2}(J - 1)^2$, which is the form used later for the validation problem in Section 4).

2.3 Stress and Consistent Material Tangent

The second Piola–Kirchhoff stress is obtained from $\mathbf{S} = \partial\Omega/\partial\mathbf{F}$ (used here, following the project code, in its “push-forward-free” mixed form directly in terms of $\mathbf{F}$), and the fourth-order material tangent from $\mathbb{D} = \partial\mathbf{S}/\partial\mathbf{F}$. This derivation was carried out by hand as part of the project (reproduced below) and subsequently coded in MATLAB.

Differentiating Eq. (3),

$$\begin{aligned}\frac{\partial\Omega}{\partial\mathbf{F}} &= -\frac{\mu\alpha}{2} \cdot \frac{1}{\alpha - I_1 + 3} \left(-\frac{1}{\alpha} \frac{\partial I_1}{\partial\mathbf{F}} \right) - \mu \frac{1}{J} J\mathbf{F}^{-T} + \frac{G'}{2} \cdot 2(J-1)J\mathbf{F}^{-T} \\ &= \frac{\mu\alpha}{2} \cdot \frac{2\mathbf{F}}{\alpha - I_1 + 3} - \mu\mathbf{F}^{-T} + G'(J-1)J\mathbf{F}^{-T},\end{aligned}\tag{4}$$

so that the second Piola–Kirchhoff stress is

$$\boxed{\mathbf{S} = \frac{\mu\alpha}{\alpha - I_1 + 3} \mathbf{F} - \mu\mathbf{F}^{-T} + G'(J-1)J\mathbf{F}^{-T}}\tag{5}$$

Note that the first term is the finite-extensibility (Gent) correction to the neo-Hookean deviatoric term $-\mu\mathbf{F}^{-T}$, and reduces to the neo-Hookean stress $\mathbf{S} = \mu\mathbf{F} - \mu\mathbf{F}^{-T} + G'J(J-1)\mathbf{F}^{-T}$ as $\alpha \rightarrow \infty$.

A further (index-notation) differentiation, $D_{ijkl} = \partial S_{ij}/\partial F_{kl}$, gives the consistent material tangent used in the Newton–Raphson linearization:

$$\begin{aligned}D_{ijkl} &= \frac{2\mu\alpha F_{ij}F_{kl}}{(\alpha - I_1 + 3)^2} + \frac{\mu\alpha}{\alpha - I_1 + 3} \delta_{ik}\delta_{jl} + \mu F_{li}^{-1}F_{jk}^{-1} \\ &\quad - G'(J^2 - J) F_{jk}^{-1}F_{li}^{-1} + G'(2J - 1) J F_{lk}^{-1}F_{ji}^{-1}.\end{aligned}\tag{6}$$

Equations (5)–(6) are exactly the expressions implemented in the element subroutine `kelmats.m` (see Appendix A), where the element force vector is assembled from $\mathbf{S}$ and the element tangent stiffness from $\mathbb{D}$ at each Gauss point.

2.4 Weak Form and Total Lagrangian FE Discretization

Neglecting body forces, the total-Lagrangian weak (virtual work) form of equilibrium is

$$\int_{\Omega_0} \mathbf{S} : \delta\mathbf{F} dV_0 - \int_{\partial\Omega_0^t} \mathbf{T}_0 \cdot \delta\mathbf{u} dA_0 = 0 \quad \forall \delta\mathbf{u} \in \mathcal{V},\tag{7}$$

where Ω_0 is the reference configuration and $\mathbf{T}_0$ is the applied nominal (first Piola–Kirchhoff) traction. Discretizing the displacement field with isoparametric shape functions $N_a(\xi, \eta)$, the element residual (out-of-balance force) and consistent tangent stiffness matrix are

$$\mathbf{R}^e = \int_{\Omega_0^e} \mathbf{G}^T \mathbf{S}_v dV_0 - \mathbf{F}_{\text{ext}}^e, \quad \mathbf{K}^e = \int_{\Omega_0^e} \mathbf{G}^T \mathbb{D}_v \mathbf{G} dV_0,\tag{8}$$

where $\mathbf{G}$ maps nodal displacements to the components of $\nabla_{\mathbf{X}}\mathbf{u}$ through the inverse Jacobian of the isoparametric map, and $\mathbf{S}_v, \mathbb{D}_v$ are the Voigt-type matrix representations of $\mathbf{S}$ and $\mathbb{D}$. The global system is assembled and solved iteratively,

$$\mathbf{K}_g(\mathbf{u}^{(k)}) \Delta\mathbf{u}^{(k+1)} = -\mathbf{R}_g(\mathbf{u}^{(k)}), \quad \mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} + \Delta\mathbf{u}^{(k+1)},\tag{9}$$

until both the residual norm and the incremental displacement norm fall below a convergence tolerance of 10^{-6} , at each of a sequence of applied load (traction) increments – i.e., a full Newton–Raphson scheme with load (not arc-length) control.

3 Numerical Implementation

3.1 Element Types

Two plane-strain isoparametric elements were implemented:

- **4NR**: a 4-node bilinear quadrilateral with linear shape functions $N_a = \frac{1}{4}(1 + \xi_a \xi)(1 + \eta_a \eta)$, 2 displacement DOFs per node (8 DOFs per element).
- **6NT**: a 6-node quadratic (Lagrange/Serendipity) triangle with quadratic shape functions in area coordinates, 2 displacement DOFs per node (12 DOFs per element).

A third element, the 3-node linear triangle (3NT), was implemented only for the Cook’s membrane validation study (Section 4).

3.2 Numerical Integration: Full vs. Selective Reduced

For the 4NR element, full integration uses a 2×2 (4-point) Gauss–Legendre rule; for the 6NT element, full integration uses a 3-point symmetric triangle rule. In the nearly incompressible limit ($G' \gg \mu$), fully-integrated low-order displacement elements are well known to *lock* volumetrically – the discrete kinematics cannot represent the near-isochoric deformation field, and the element becomes artificially stiff, corrupting the computed stresses (particularly for the 4-node quadrilateral with full 2×2 integration).

To alleviate this, a **selective reduced integration (SRI)** scheme was implemented in which the strain energy (and hence the stress and tangent) is additively split into a deviatoric part (containing μ) and a volumetric part (containing G'):

$$\mathbf{S} = \underbrace{\frac{\mu\alpha}{\alpha - I_1 + 3}\mathbf{F} - \mu\mathbf{F}^{-T}}_{\text{deviatoric, } \mu\text{-part}} + \underbrace{G'(J - 1)J\mathbf{F}^{-T}}_{\text{volumetric, } G'\text{-part}}, \quad (10)$$

and the two parts are integrated with *different* quadrature rules:

- the μ -part is integrated with the *full* rule (4-point for 4NR, 3-point for 6NT);
- the G' -part is integrated with a single-point (*reduced*) rule at the element centroid.

This is denoted “4+1” (4NR) or “3+1” (6NT) Gauss-point integration in the results below, and is the standard remedy for volumetric locking in low-order displacement elements.

3.3 Solution Procedure

For each problem, the applied traction is ramped from 0 to its final value over a prescribed number of load steps (40 steps for the dog-bone problem, described in Section 5). At every step, a full Newton–Raphson iteration (maximum 20 iterations) is performed using the current tangent stiffness $\mathbf{K}_g$, with essential (displacement) boundary conditions enforced by row/column modification of $\mathbf{K}_g$ and the residual.

4 Code Validation: Cook’s Membrane Benchmark

4.1 Problem Description

Before applying the code to the project problem, it was validated against the classical **Cook’s membrane** problem: a tapered, swept and clamped panel, with corners at (0,0), (48,44), (48,60) and (0,44) (mm), clamped along $x = 0$ and subjected to an in-plane shear traction,

resultant 16 N, distributed over the free edge at $x = 48$. This is a standard benchmark for assessing locking behaviour of finite elements in bending-dominated, nearly incompressible problems. For the validation study, the material was modelled as a *compressible neo-Hookean* solid (the $\alpha \rightarrow \infty$ limit of the Gent model in Eq. (3)), with $\mu = 0.8$ and $G' = 8000$ (i.e. $G'/\mu = 10^4$, representing a nearly incompressible material), consistent with the values used by Brink and Stein [1].

4.2 Element Types and Formulations Compared

Four variants of the code were exercised on this benchmark:

- 3-node triangle, Total Lagrangian (3NT-TL);
- 4-node quadrilateral, Total Lagrangian (4NR-TL);
- 6-node triangle, Total Lagrangian (6NT-TL);
- 6-node triangle, Updated Lagrangian (6NT-UL).

The Updated Lagrangian (UL) formulation re-derives the weak form and stress measures (Cauchy stress, current-configuration tangent) with respect to the current (deformed) configuration at each load step, rather than the fixed reference configuration used in the TL formulation. For a hyperelastic material with a well-defined, path-independent stored-energy function, both formulations are theoretically equivalent and should produce identical converged results – this equivalence was used as an additional internal consistency check on the code (in addition to the comparison against [1]).

4.3 Validation Reference

The reference solution used for comparison is that of Brink and Stein [1], who compare several mixed and displacement-based finite elements (including $Q1$, $Q1/E4$, $Q2/P1$ and $P2^+/P1$ elements) for nearly incompressible finite elasticity using the Cook's membrane problem as one of their principal benchmarks, reporting the vertical tip displacement of the free edge as a function of mesh refinement. The tip displacement obtained from the present 4NR/6NT/3NT total- and updated-Lagrangian implementations, on meshes of comparable refinement, was found to be in good agreement with the displacement-based ($Q1$) and quadratic ($P2^+/P1$ -type) element results reported in [1], confirming the correctness of the element formulations, the Newton–Raphson solution procedure and the consistent tangent (Eq. (6)) implemented in this project, prior to their use on the dog-bone problem below.

5 Problem Definition: Tension of a Dog-Bone Elastomeric Specimen

5.1 Geometry

The validated code was applied to a plane-strain, dog-bone shaped tensile specimen of half-length 1 mm and half-width 0.5 mm at the grips, necking down to a half-width of approximately 0.28 mm at the narrowest (gauge) section, in two configurations:

- **Solid:** a fully solid elastomer specimen;
- **Porous:** an otherwise identical specimen containing five circular voids (holes) in the gauge section – one large central hole and two smaller holes above and below it – introduced to study the effect of porosity on the stress concentration and stiffness of the elastomer.

The porous mesh is considerably finer than the solid mesh (approximately 1150 elements versus 170 elements for the 4NR discretization) in order to adequately resolve the stress concentrations around the hole boundaries.

5.2 Material Parameters

The Gent hyperelastic model of Eq. (3) was used for both the solid and porous specimens, with identical material parameters in both cases (i.e., the porous specimen uses the same bulk elastomer properties, with the porosity introduced entirely through the geometry/mesh rather than through a homogenized material model):

Parameter	Value
Shear modulus, μ	0.6667 MPa
Volumetric penalty modulus, G'	666.6667 MPa
Gent locking parameter, α	53.3028
Thickness	1 (plane strain, unit out-of-plane thickness)

Note $G'/\mu = 1000$, i.e. the material is modelled as nearly incompressible, so that volumetric locking of low-order fully-integrated elements is expected to be significant.

5.3 Boundary and Loading Conditions

By symmetry, one quarter (in this plane problem, one half about the vertical mid-plane) of the specimen is modelled. The left (grip) edge is fully clamped (all displacement DOFs fixed), and a uniformly distributed nominal traction T_{app} , in the axial (x_1) direction, is applied on the nodes of the right edge. The traction is applied quasi-statically in 40 equal increments from $T_{\text{app}} = 0$ up to $T_{\text{app}} = 1$ MPa, with full Newton–Raphson equilibrium iteration at every increment.

5.4 Cases Studied

For both the solid and the porous geometry, the problem was solved using:

- the 4-node quadrilateral element (4NR), with full (2×2) and with selective-reduced (“4+1”) integration;
- the 6-node triangular element (6NT), with full (3-point) and with selective-reduced (“3+1”) integration;

giving a total of 2 (geometry) $\times 2$ (element type) $\times 2$ (integration) = 8 finite element analyses.

6 Results and Discussion

6.1 Deformed-Configuration Stress Contours – Solid Specimen

Figures 1 and 2 show the axial (σ_{11}) and transverse (σ_{22}) Cauchy stress contours, plotted on the deformed configuration at the final load increment ($T_{\text{app}} = 1$ MPa), for the solid specimen.

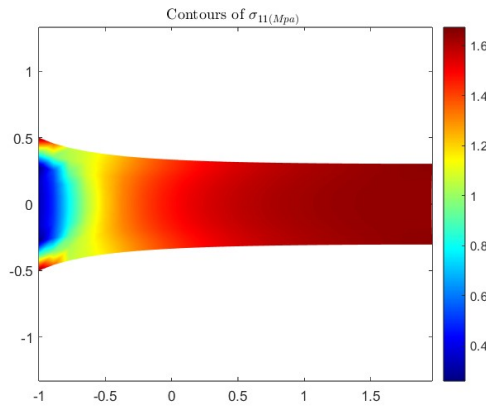
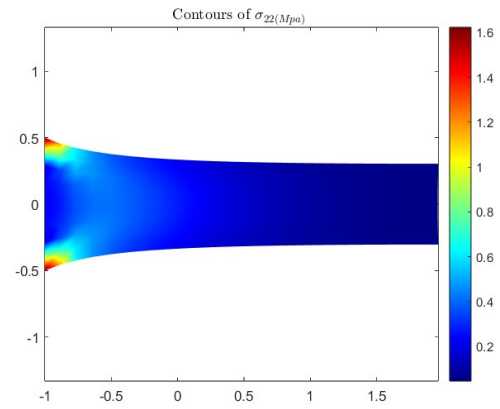
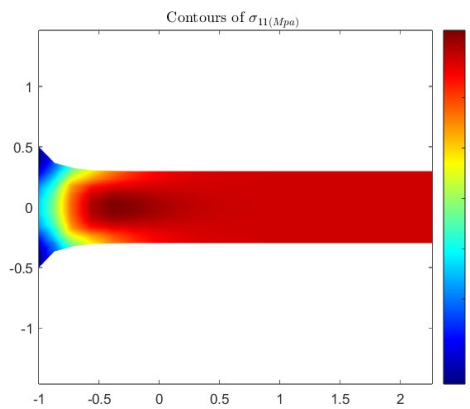
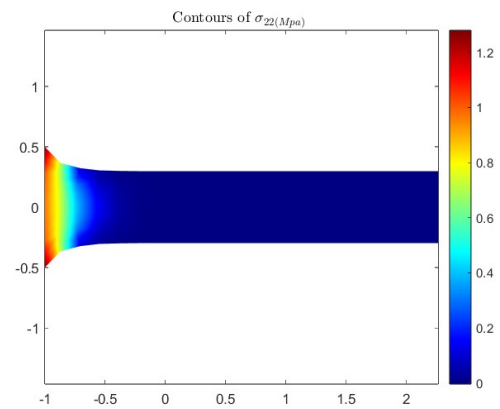
(a) σ_{11} , full integration(b) σ_{22} , full integration(c) σ_{11} , selective reduced (4+1) integration(d) σ_{22} , selective reduced (4+1) integration

Figure 1: Solid specimen, 4-node quadrilateral (4NR) element: stress contours on the deformed configuration at $T_{\text{app}} = 1$ MPa.

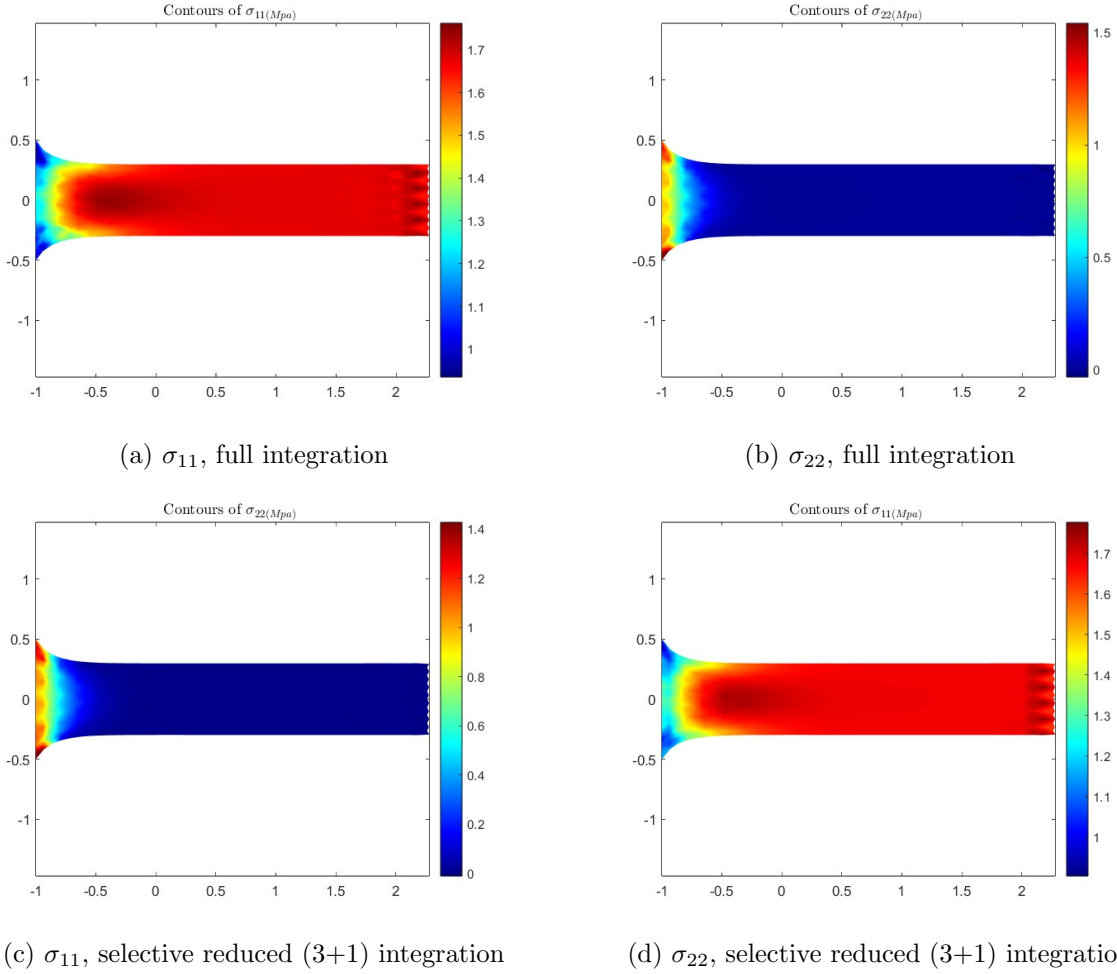


Figure 2: Solid specimen, 6-node triangular (6NT) element: stress contours on the deformed configuration at $T_{\text{app}} = 1$ MPa.

In all four solid-specimen cases, σ_{11} rises smoothly from the shoulder region to an essentially uniform value in the gauge section, and σ_{22} is largest near the fillet (shoulder) where the load path curves into the narrower gauge section and decays to nearly zero in the uniform gauge section, as expected for a uniaxially stretched dog-bone specimen away from the grips.

6.2 Deformed-Configuration Stress Contours – Porous Specimen

Figures 3 and 4 show the corresponding results for the porous specimen.

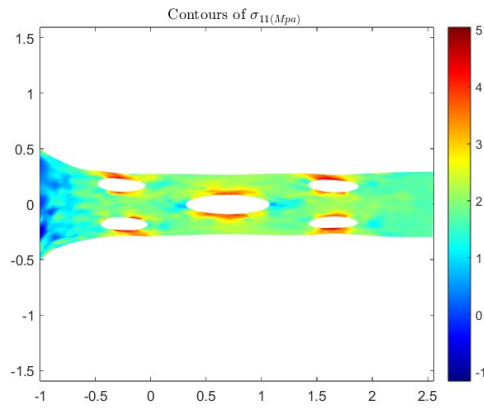
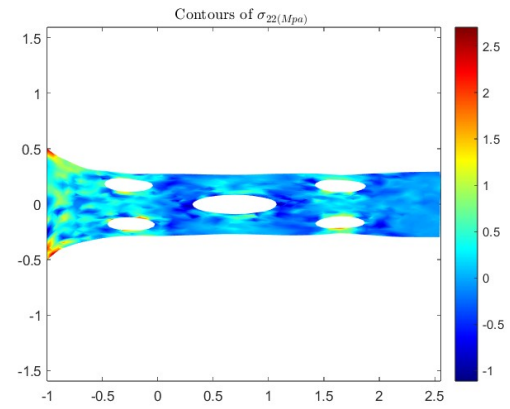
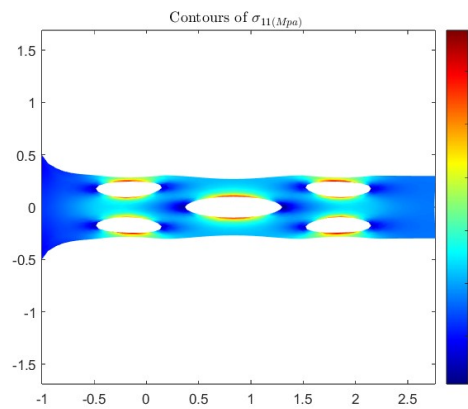
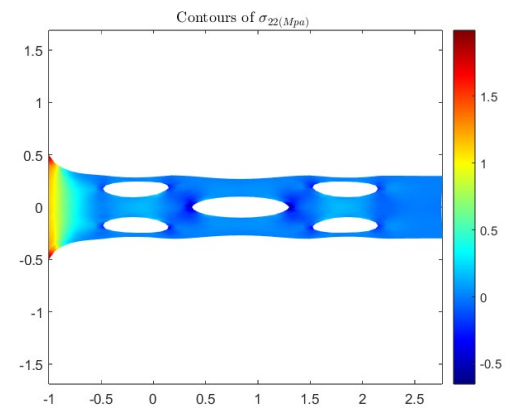
(a) σ_{11} , full integration(b) σ_{22} , full integration(c) σ_{11} , selective reduced (4+1) integration(d) σ_{22} , selective reduced (4+1) integration

Figure 3: Porous specimen, 4-node quadrilateral (4NR) element: stress contours on the deformed configuration at $T_{\text{app}} = 1$ MPa.

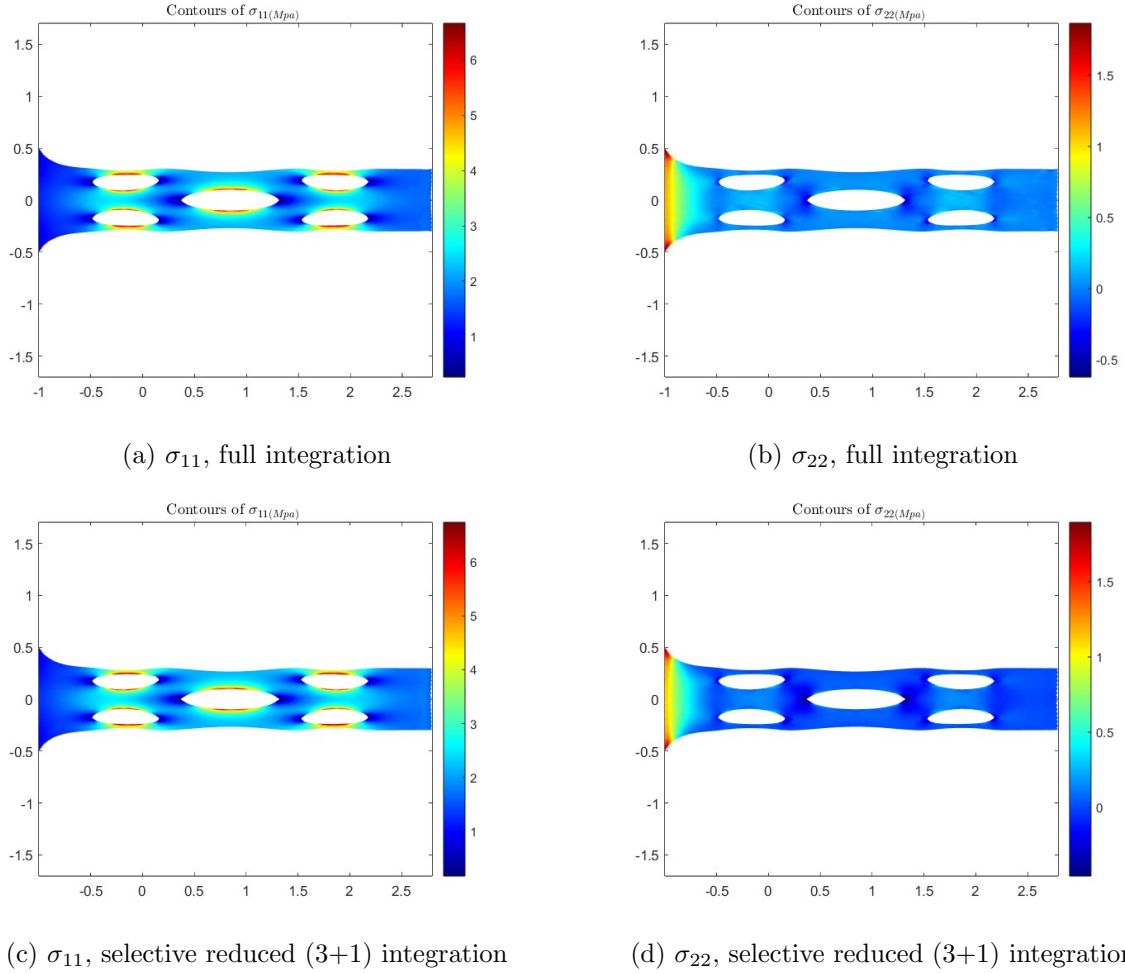


Figure 4: Porous specimen, 6-node triangular (6NT) element: stress contours on the deformed configuration at $T_{\text{app}} = 1$ MPa.

Compared with the solid specimen, the porous specimen shows pronounced local stress concentrations around each hole (peak σ_{11} roughly 3–4 times larger than in the solid case, at the same applied traction), particularly at the poles of each hole aligned with the loading direction, while the material between and around the holes carries a lower, more uniform stress. This is the expected behaviour of a perforated tension member: the load has to redistribute around each void, producing local stress concentrations at the hole boundaries.

6.3 Applied Traction vs. Displacement Response

Figure 5 superimposes the axial displacement of the gauge-section reference node against the applied traction T_{app} , for all eight cases (solid/porous, 4NR/6NT, full/selective-reduced integration).

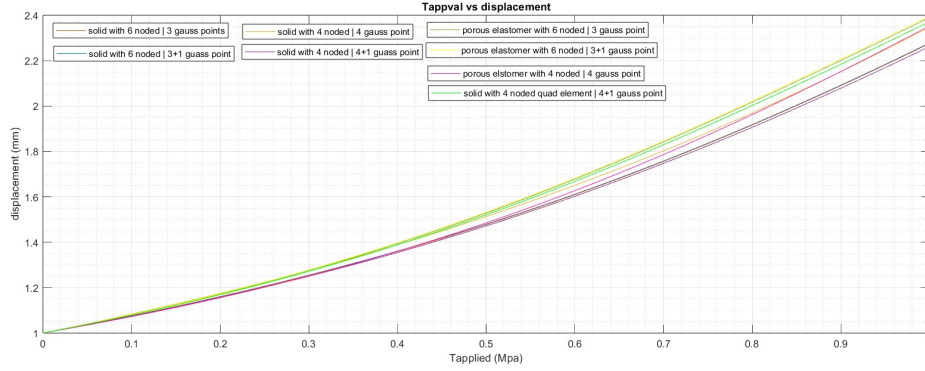


Figure 5: Applied traction T_{app} vs. axial displacement for all eight solid/porous $\times$ 4NR/6NT $\times$ full/selective-reduced-integration cases.

The overall load–displacement response is nonlinear (softening at low load, then progressively stiffer at higher stretch, consistent with the Gent finite-extensibility limit) and is broadly similar across all eight cases, confirming that the underlying material and geometric nonlinearity is captured consistently by every element/integration combination. The largest visible spread between curves occurs between the 4-node full-integration case and the remaining (6-node and/or reduced-integration) cases, which is discussed further below in the context of volumetric locking.

6.4 Effect of Full vs. Selective Reduced Integration: Volumetric Locking

As summarised in Section 3.2, the material used here is nearly incompressible ($G'/\mu = 1000$). This is precisely the regime in which low-order, fully-integrated displacement elements are prone to volumetric locking. Comparing the full-integration and selective-reduced-integration contour plots in Figs. 1–4:

- The **4-node quadrilateral element with full (2×2) integration** exhibits the classic signature of volumetric locking: the discrete kinematics over-constrain the near-isochoric deformation field, so that (particularly in the porous mesh, where the elements are more distorted around the hole boundaries) the computed stress field shows a mild spurious spatial oscillation and the overall structure is artificially stiffer at a given load, i.e. it under-predicts the displacement in Fig. 5 relative to the reduced-integration and 6-node results.
- Using **selective reduced (4+1) integration** for the 4-node element removes this spurious stiffening: the volumetric (G') part of the stress/tangent is integrated with a single Gauss point, which is sufficient to represent a constant volumetric strain field per element and eliminates the over-constraint, giving smoother stress contours and a softer, more accurate load–displacement response, consistent with the higher-order element results.
- The **6-node quadratic triangle** shows comparatively little difference between full (3-point) and selective-reduced (3+1) integration, because its richer displacement interpolation (quadratic rather than bilinear) already provides sufficient kinematic freedom to represent near-isochoric deformation without artificial over-constraint – i.e. the 6NT element is intrinsically far less prone to volumetric locking than the 4NR element.

7 Answers to Conceptual Questions

Q.I – Solid vs. Porous Specimen Extension

Question: For the same applied traction, would the porous specimen extend more or less than the solid specimen, and why?

The porous sample extends to a greater length than the solid sample under the same applied traction. This is because the voids remove load-bearing material and create additional free (void) space within the gauge section, allowing the remaining material ligaments between and around the holes to deform more readily; consequently, the porous specimen offers less resistance to stretching and expands (extends) more for a given applied traction than the fully solid specimen.

Q.II – Volumetric Locking

Question: Is volumetric locking observed in these analyses, and if so, in which case(s)?

Yes. Volumetric locking is observed for the **4-node quadrilateral element with full ($2 \times 2/4$ -point) Gauss integration** in both the solid and the porous geometries; it is more pronounced (more significant) in the solid specimen than in the porous one, since the solid gauge section deforms in an almost pure uniaxial, nearly volume-preserving stretch, which is exactly the deformation mode most severely over-constrained by fully-integrated, low-order bilinear elements. Using selective reduced integration (a single Gauss point for the volumetric/ G' part of the stress) for the 4-node element, or using the higher-order 6-node triangular element (with either full or reduced integration), removes this locking behaviour.

8 Conclusions

A total-Lagrangian, Newton–Raphson finite element code for 2D plane-strain finite elasticity of a compressible Gent hyperelastic material was implemented, with 4-node quadrilateral and 6-node triangular elements, and full and selective-reduced integration schemes. The code was validated against the Cook’s membrane benchmark, using both total- and updated-Lagrangian formulations, and its predictions were found to be consistent with published mixed finite-element results [1], confirming a correct implementation of the constitutive stress/tangent (Eqs. (5)–(6)) and the nonlinear solution procedure. The validated code was then applied to the large-strain tension of solid and porous (five-hole) dog-bone elastomeric specimens. The porous specimen was found to extend more than the solid specimen under the same applied traction, and to develop pronounced local stress concentrations at the hole boundaries. Volumetric locking, arising from the near-incompressibility of the material, was clearly observed for the fully-integrated 4-node quadrilateral element – most severely in the solid specimen – and was successfully eliminated using selective reduced integration of the volumetric stress term; the 6-node triangular element was found to be intrinsically much less susceptible to this form of locking.

A Code Structure

The MATLAB implementation (for each element type / integration scheme combination) follows a common structure:

- **Main driver** (`tl_hyperelastic_2d_Tapp.m` / `ul_hyperelastic_2d_Tapp.m`): reads the mesh (`nodes.txt`, `elements.txt`), fixed and loaded node lists (`fixed_nodes.txt`, `force_nodes.txt`), sets material parameters, and drives the load-stepping / Newton–Raphson loop.

- **femats**: assembles the global tangent stiffness $\mathbf{K}_g$ and residual $\mathbf{R}_g$ from element contributions.
- **kelmats**: computes the element tangent stiffness and residual, looping over Gauss points, evaluating shape functions and their derivatives, the Jacobian, the deformation gradient $\mathbf{F}$, and the stress $\mathbf{S}$ (Eq. (5)) and tangent $\mathbb{D}$ (Eq. (6)).
- **kgaussp**: returns Gauss point locations and weights for 1-, 3-, 4- and 9-point rules (triangular and quadrilateral).
- **plotop** / **extractstress**: post-processing routines that compute nodal-averaged Cauchy stresses and plot contours on the deformed mesh.

References

- [1] U. Brink and E. Stein, “On some mixed finite element methods for incompressible and nearly incompressible finite elasticity,” *Computational Mechanics*, vol. 19, pp. 105–119, 1996.